

Top 10 Features of SQL Server

By Don Schlichting

Introduction

Microsoft SQL Server 2008 R2 is the latest release of SQL Server. This article will introduce the top 10 features and benefits of SQL Server 2008 R2. The "R2" tag indicates this is an intermediate release of SQL Server and not a major revision. However, there are a number of interesting new features for both DBAs and developers alike. At the time of this article, R2 is available as a CTP (Community Technology Preview). In addition to new features, there are two new editions as well, SQL Server 2008 R2 Datacenter and SQL Server 2008 R2 Parallel Data Warehouse.

Report Builder 3.0

Report Builder is a tool set for developing rich reports that can be delivered over the web. Some of the features of Report Builder include the ability to create reports containing graphs, charts, tables, and printing controls. In addition, Report Builder also supports drill downs and sorting. If you are familiar with the third party tool Crystal Reports, then you have good idea of what to expect from Report Builder.

New features in SQL Server / Report Builder 3.0 include: Map Layers, which can hold spatial and analytical data and will integrate with Microsoft Virtual Earth. Indicators, these are gauges used to show the state of one value. Report Parts, this object can be reused or shared between multiple reports. Aggregate Calculating, this allows you to calculate the total value of other aggregate calculated totals.

SQL Server Datacenter

The new Datacenter edition of SQL Server is targeted towards Enterprise Edition users who require a greater performance platform. The new edition will support 256 logical processors, high numbers of instances, and as much memory as the operating system will support.

SQL Server Parallel Data Warehouse

Another new SQL Server edition, Parallel Data Warehouse, formally codenamed "Madison", specializes in handling extremely large amounts of data. This new version uses massively parallel processing to spread large tables over multiple SQL nodes. The multiple nodes are handled by a propriety Microsoft technology called Ultra Shared Nothing. This new technology is described as a Control Node spreading queries to Computer Nodes, evenly distributed, then collecting the results.

StreamInsight

New in SQL Server 2008 R2 is component called StreamInsight. This interesting component allows streaming data to be analyzed on the fly. Meaning the data is processed directly from the source stream prior to being saved in a SQL Server table. This could be extremely handy if you're running a real time system and need to analyze data but can't afford the latency of a committed write to a table first. Examples usually cited for this application

include stock trading streams, click stream web analytics, and industrial process controls. Multiple input streams can be simultaneously monitored.

Master Data Services

Master Data Services (MDS) is both a concept and a product. The concept of a Master Data Service is that there is a central data gate keeper of core business data. Data items such as customer billing addresses, employee/customer names, and product names should be centrally managed so that all consuming applications have the same information. The Microsoft example given is a company that has a customer address record in the customer table but a different address in the mailing table. A Master Data Service application would ensure that all tables would have only one correct address. While an MDS can be a homegrown application, SQL Server 2008 R2 includes an application and an interface to manage the central data.

PowerPivot for SharePoint

PowerPivot is an end-user tool that works in conjunction with SharePoint, SQL Server, and Excel 2010 to process large amounts of data in seconds. PowerPivot works like an Excel Pivot Table, and includes analytical capabilities.

Data-Tier Application

A Data-Tier Application (abbreviated as DAC –no idea what the C stands for, and not to be confused with the Windows Data Access Components also abbreviated as DAC) is an object that stores all the needed database information for a project, such as login, tables, and procedures into one package that can be consumed by Visual Studio. By creating a Data-Tier Application, a SQL Server package version could be saved with each Visual Studio build of your application. This would allow application code builds to be married to a database build in an easily managed way.

Unicode Compression

SQL Server uses a new algorithm known as Simple Compression Scheme for Unicode storage. This reduces the amount of disk space used by Unicode characters. This new format happens automatically and is managed by the SQL Server engine so no programming changes are required of the DBA.

SQL Server Utility

The new SQL Server Utility is a repository object for centrally controlling multiple SQL Server instances. Performance data and configuration policies can be stored in a single Utility. The Utility also includes an Explorer tool where multi-server dashboards can be created.

Multi Server Dashboards

While the SQL Server Management Studio could always connect to multiple servers, each was managed independently with no central view of all of them. Now with SQL Server, Dashboards showing combined server data can be created.

Conclusion

SQL Server 2008 R2 is the latest release of Microsoft SQL Server. The "R2" tag indicates this is an intermediate release of SQL Server and not a major revision. There are a number of compelling features in this version for both developers and DBAs alike. Here are the Top 10 new features in SQL Server 2008 R2:

1. Report Builder 3.0
2. SQL Server Datacenter
3. SQL Server Parallel Data Warehouse
4. StreamInsight
5. Master Data Services
6. PowerPivot for SharePoint
7. Data-Tier Application
8. Unicode Compression
9. SQL Server Utility
10. Multi Server Dashboards